Securing understanding of factors, multiples, squares and cubes

Task A: Multiplication tables to identify multiples

1) Here is a 6 by 6 multiplication table. Answer true or false to each statement providing a justification for your answer.

✖	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

Statement	True	False	Justification
a) 12 is a factor of 4		✓	12 is a multiple of 4 or 4 is a factor of 12
b) 36 is a multiple of 3	✓		$3 \times 12 = 36$ or 36 is a multiple of 3
c) All the multiples of 6 can be seen in this table		✓	There are an infinite number of multiples

2) Using a 12 by 12 multiplication table:

✖	1	2	3	4	5	6	7	8	9	10	11	12
1	1	2	3	4	5	6	7	8	9	10	11	12
2	2	4	6	8	10	12	14	16	18	20	22	24
3	3	6	9	12	15	18	21	24	27	30	33	36
4	4	8	12	16	20	24	28	32	36	40	44	48
5	5	10	15	20	25	30	35	40	45	50	55	60
6	6	12	18	24	30	36	42	48	54	60	66	72
7	7	14	21	28	35	42	49	56	63	70	77	84
8	8	16	24	32	40	48	56	64	72	80	88	96
9	9	18	27	36	45	54	63	72	81	90	99	108
10	10	20	30	40	50	60	70	80	90	100	110	120
11	11	22	33	44	55	66	77	88	99	110	121	132
12	12	24	36	48	60	72	84	96	108	120	132	144

a) Find the largest multiple of 7 84

b) Find the largest multiple of 3 144

c) Find the largest multiple of 5 120

d) Find the lowest multiple of 3 and 5 15

Task B: Multiplication tables to identify factors

1) Using an 8 by 8 multiplication table, find:

a) all the factors of 8

1,8,2 and 4

b) all the factors of 5

1, 2 and 5

c) all the factors of 4

1, 2 and 4

d) all the factors of 7

1 and 7

e) What is their common factor?

1

✖	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	4	6	8	10	12	14	16
3	3	6	9	12	15	18	21	24
4	4	8	12	16	20	24	28	32
5	5	10	15	20	25	30	35	40
6	6	12	18	24	30	36	42	48
7	7	14	21	28	35	42	49	56
8	8	16	24	32	40	48	56	64

2) Using a 6 by 6 multiplication table, answer true or false to each statement providing a justification for your answer.

✖	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

Statement	True	False	Justify
a) 4 only has three factors	✓		1,2 and 4
b) All the factors of 6 are seen	✓		1, 6, 2 and 3
c) 3 has an infinite number of factors		✓	1 and 3
d) There are five square numbers seen		✓	1, 4, 9, 16, 25 and 36

3) What number is remaining when you eliminate:

a) all the multiples of 5?

b) all the factors of 24?

c) all the multiples of 3?

d) all the multiples of 4?

e) all the factors of 14?

×	1	2	3	4	5	6	7	8	9	10
1	~~1~~	~~2~~	~~3~~	~~4~~	~~5~~	~~6~~	~~7~~	~~8~~	~~9~~	~~10~~
2	~~2~~	~~4~~	~~6~~	~~8~~	~~10~~	~~12~~	~~14~~	~~16~~	~~18~~	~~20~~
3	~~3~~	~~6~~	~~9~~	~~12~~	~~15~~	~~18~~	~~21~~	~~24~~	~~27~~	~~30~~
4	~~4~~	~~8~~	~~12~~	~~16~~	~~20~~	~~24~~	~~28~~	~~32~~	~~36~~	~~40~~
5	~~5~~	~~10~~	~~15~~	~~20~~	~~25~~	~~30~~	~~35~~	~~40~~	~~45~~	~~50~~
6	~~6~~	~~12~~	~~18~~	~~24~~	~~30~~	~~36~~	~~42~~	~~48~~	~~54~~	~~60~~
7	~~7~~	~~14~~	~~21~~	~~28~~	~~35~~	~~42~~	49	~~56~~	~~63~~	~~70~~
8	~~8~~	~~16~~	~~24~~	~~32~~	~~40~~	~~48~~	~~56~~	~~64~~	~~72~~	~~80~~
9	~~9~~	~~18~~	~~27~~	~~36~~	~~45~~	~~54~~	~~63~~	~~72~~	~~81~~	~~90~~
10	~~10~~	~~20~~	~~30~~	~~40~~	~~50~~	~~60~~	~~70~~	~~80~~	~~90~~	~~100~~